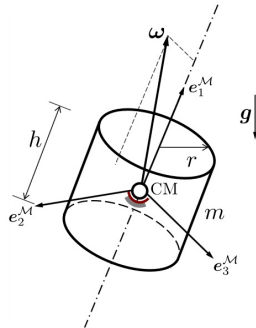


Problems Week 10

Problem 1: Rotating Cylinder

A uniform cylinder of mass m , height h , radius r and principal moment of inertia tensor $\mathbf{I}_{CM}$ is hinged at its center of mass CM. It is given an initial angular velocity $\boldsymbol{\omega}$, which lies in the $\mathbf{e}_1^M$ - $\mathbf{e}_3^M$ -plane of a principal coordinate frame rotating with the cylinder, as shown below. Gravity (acting downwards) is the only force acting on the body.

Given: $m, \omega_1, \omega_3, [\mathbf{I}_{CM}]_M = \begin{bmatrix} \frac{1}{2}mr^2 & 0 & 0 \\ 0 & \frac{m}{12}(3r^2 + h^2) & 0 \\ 0 & 0 & \frac{m}{12}(3r^2 + h^2) \end{bmatrix}, g$



What is the ratio h/r that keeps the angular velocity vector in the $\mathbf{e}_1^M$ - $\mathbf{e}_3^M$ -plane?

$\mathcal{L}^M = \omega$ & $\mathbf{I}_{CM}$ is diag $\Rightarrow$ Euler eq.

$$\begin{aligned} \hat{I}_1 \dot{\omega}_1 + (\hat{I}_3 - \hat{I}_2) \omega_3 \omega_2 &= [M_{B,1}]_{\hat{M}} \\ (2) \quad \hat{I}_2 \dot{\omega}_2 + (\hat{I}_1 - \hat{I}_3) \omega_1 \omega_3 &= [M_{B,2}]_{\hat{M}} \\ \hat{I}_3 \dot{\omega}_3 + (\hat{I}_2 - \hat{I}_1) \omega_2 \omega_1 &= [M_{B,3}]_{\hat{M}} \end{aligned}$$

here $\omega_2 = 0$ & $\dot{\omega}_2 = 0$

(2) $(\hat{I}_1 - \hat{I}_3) \omega_1 \omega_3 = 0$ ← since no external torque

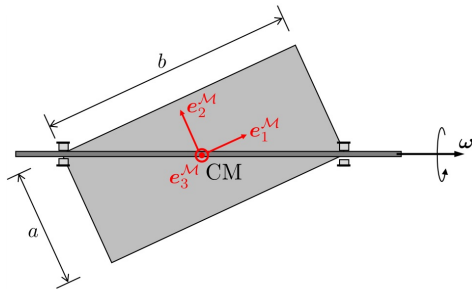
$$\frac{1}{2}mr^2 - \frac{m}{12}(3r^2 + h^2) = 0$$

↳ do the math $\Rightarrow \underline{\underline{\frac{h}{r} = \sqrt{3}}}$

Problem 2: Rotating Plate

A thin, homogeneous, rectangular plate of mass M and side lengths a and b rotates around its diagonal with a constant angular velocity ω , as indicated below. Its principal moment of inertia tensor in the given frame $\mathcal{M}$ (rotating with the plate) is denoted by $[\mathbf{I}_{CM}]_{\mathcal{M}}$. Neglect gravity.

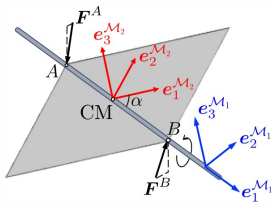
Given: $M, a, b, \omega, [\mathbf{I}_{CM}]_{\mathcal{M}} = \frac{1}{12}M \begin{bmatrix} a^2 & 0 & 0 \\ 0 & b^2 & 0 \\ 0 & 0 & (a^2 + b^2) \end{bmatrix}$



- (a) What is the magnitude F of the force $\mathbf{F}$ in the bearings? (b) What is the kinetic energy of the plate?

$$\mathbf{F}^A = F_2^A \mathbf{e}_2^{M_1} + F_3^A \mathbf{e}_3^{M_1}, \quad \mathbf{F}^B = F_2^B \mathbf{e}_2^{M_1} + F_3^B \mathbf{e}_3^{M_1}.$$

→ forces of support only
in radial direction



M_1 rotates with shaft: $\underline{\Omega}^{M_1} = \omega$

M_2 " " " & is aligned with princ. axes (i.e. in M_2 frame)

① LMB: $m \cdot \ddot{\mathbf{y}}_{CM} = \mathbf{F}^A + \mathbf{F}^B \Rightarrow [\mathbf{F}_2^A]_{M_1} = -[\mathbf{F}_2^B]_{M_1} \quad \parallel \quad [\mathbf{F}_3^A]_{M_1} = -[\mathbf{F}_3^B]_{M_1}$

We want to use M_2 frame since we can use Euler eq. & have $[\hat{\mathbf{I}}_{CM}]_{M_2}$

↳ (i) find torques of $\mathbf{F}^{ext}$ w.r.t. the CM (we can do this in M_1)

$$\begin{aligned} \underline{M}_{CM} &= \underline{r}_{CA} \times \mathbf{F}^A + \underline{r}_{CB} \times \mathbf{F}^B \\ &= \begin{pmatrix} -\frac{\sqrt{a^2+b^2}}{2} \\ 0 \\ 0 \end{pmatrix}_{M_1} \times \begin{pmatrix} 0 \\ F_2^A \\ F_3^A \end{pmatrix}_{M_1} + \begin{pmatrix} \frac{\sqrt{a^2+b^2}}{2} \\ 0 \\ 0 \end{pmatrix}_{M_1} \times \begin{pmatrix} 0 \\ F_2^B \\ F_3^B \end{pmatrix}_{M_1} = \sqrt{a^2+b^2} \left(F_3^A \mathbf{e}_2^{M_1} - F_2^A \mathbf{e}_3^{M_1} \right) \end{aligned}$$

plugging in $[\mathbf{F}^A] \rightarrow -[\mathbf{F}^B]$

(ii) $\underline{\omega}^{M_2} = \underline{\underline{R}}^{M_2 M_1} \omega \mathbf{e}_1^{M_1} = \omega \cos(\alpha) \mathbf{e}_1^{M_2} - \omega \sin(\alpha) \mathbf{e}_2^{M_2}$

(iii) $[\mathbf{M}_{CM}]_{M_2} = \sqrt{a^2+b^2} \cdot \begin{pmatrix} F_3^A \sin \alpha \\ F_3^A \cos \alpha \\ -F_2^A \end{pmatrix}$

$$[\underline{\underline{R}}^{M_2 M_1}]_{M_1} = \begin{pmatrix} \cos \alpha & \sin \alpha & 0 \\ -\sin \alpha & \cos \alpha & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

② Euler eq. with $\dot{\omega} = 0$ as given in Ex.

$$\begin{cases} 0 + \sqrt{a^2 + b^2} F_3^A \sin \alpha = 0 \\ 0 + \sqrt{a^2 + b^2} F_3^A \cos \alpha = 0 \\ 0 - \sqrt{a^2 + b^2} F_2^A = -\frac{1}{12} M (b^2 - a^2) \omega^2 \sin \alpha \cos \alpha \end{cases} \Rightarrow F_3^A = 0 \rightarrow \text{react. forces in plane of rectangle}$$

$$\begin{aligned} \hat{I}_1 \dot{\omega}_1 + (\hat{I}_3 - \hat{I}_2) \omega_3 \omega_2 &= [M_{B,1}]_{\mathcal{M}} \\ \hat{I}_2 \dot{\omega}_2 + (\hat{I}_1 - \hat{I}_3) \omega_1 \omega_3 &= [M_{B,2}]_{\mathcal{M}} \\ \hat{I}_3 \dot{\omega}_3 + (\hat{I}_2 - \hat{I}_1) \omega_2 \omega_1 &= [M_{B,3}]_{\mathcal{M}} \end{aligned}$$

$\hookrightarrow$ since $\sin \alpha = \frac{a}{\sqrt{a^2 + b^2}}$ & $\cos \alpha = \frac{b}{\sqrt{a^2 + b^2}}$

$$\Rightarrow F_2^A = \frac{1}{12} M \omega^2 (b^2 - a^2) \frac{ab}{(a^2 + b^2)^{3/2}} = -F_2^B$$

Since only $F_2^A \neq 0 \Rightarrow \underline{\underline{|F| = F_2^A}}$

b) $T = \frac{1}{2} \underline{\omega} \underline{I}_{cm} \underline{\omega} + \frac{1}{2} M |\underline{v}_{cm}|^2 = \frac{1}{2} (I_{11} \omega_1^2 + I_{22} \omega_2^2) + 0$ makes sense as we are in M_2 frame (we only have $[\hat{I}_{cm}]_{M_2}$ given $\ddot{\omega}$)

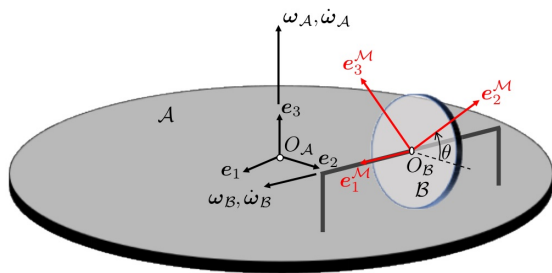
$\downarrow$ plug in

$$\underline{\underline{= \frac{1}{12} M \omega^2 \frac{a^2 b^2}{a^2 + b^2}}}$$

Problem 3: Disk on a Platform

A platform $\mathcal{A}$ is rotating about the fixed vertical axis e_3 with a given angular velocity ω_A and angular acceleration $\dot{\omega}_A$. At the same time, an internal motor spins a uniform disk $\mathcal{B}$ about its axis of rotational symmetry (aligned with e_1^M), which is rigidly mounted to the platform, with a given angular velocity ω_B and angular acceleration $\dot{\omega}_B$. The moving frame $\{e_1^M, e_2^M, e_3^M\}$ is attached to $\mathcal{B}$ and centered in the center of mass O_B . The frame $\{e_1, e_2, e_3\}$ is fixed and centered in the center O_A of $\mathcal{A}$. The principal moment of inertia tensor of the disk with respect to the $\mathcal{M}$ -frame is given by $[I_{O_B}]_{\mathcal{M}}$. Neglect gravity.

Given: $\omega_A, \dot{\omega}_A, \omega_B, \dot{\omega}_B, [I_{O_B}]_{\mathcal{M}} = \begin{bmatrix} 2I & 0 & 0 \\ 0 & I & 0 \\ 0 & 0 & I \end{bmatrix}, I$



Determine the components $[M_i^{O_B}]_{\mathcal{M}}$, i.e., the torque that is applied to the disk $\mathcal{B}$ at O_B by the spinning axle mounted on the rotating platform $\mathcal{A}$, measured in the moving frame $\mathcal{M}$.

$$\underline{\Omega}^M = \underline{\omega}_B \quad \& \quad I \text{ is diag} \Rightarrow$$

$$\left. \begin{aligned} \hat{I}_1 \dot{\omega}_1 + (\hat{I}_3 - \hat{I}_2) \omega_3 \omega_2 &= [M_{B,1}]_{\mathcal{M}} \\ \hat{I}_2 \dot{\omega}_2 + (\hat{I}_1 - \hat{I}_3) \omega_1 \omega_3 &= [M_{B,2}]_{\mathcal{M}} \\ \hat{I}_3 \dot{\omega}_3 + (\hat{I}_2 - \hat{I}_1) \omega_2 \omega_1 &= [M_{B,3}]_{\mathcal{M}} \end{aligned} \right\}$$

Euler eq.

$$\underline{\omega}_{tot} = \omega_A e_3 + \omega_B e_1^M \rightarrow \underline{\omega}^M = \begin{pmatrix} \omega_B \\ \sin \theta \omega_A \\ \cos \theta \omega_A \end{pmatrix}$$

← abs. ang. vel. represented in unit vector basis of $\mathcal{M}$ -frame

$$\underline{\dot{\omega}}^M = \begin{pmatrix} \dot{\omega}_B \\ \dot{\omega}_A \sin \theta + \omega_A \dot{\theta} \cos \theta \\ \dot{\omega}_A \cos \theta - \omega_A \dot{\theta} \sin \theta \end{pmatrix} \quad \text{since } \dot{\theta} = \omega_B$$

$$= \begin{pmatrix} \dot{\omega}_B \\ \dot{\omega}_A \sin \theta + \omega_A \omega_B \cos \theta \\ \dot{\omega}_A \cos \theta - \omega_A \omega_B \sin \theta \end{pmatrix}$$

$$\Rightarrow \text{plug into Euler: } \begin{cases} 2I \cdot \dot{\omega}_B + 0 = [M_{B,1}]_{\mathcal{M}} \\ I \cdot (\dot{\omega}_A \sin \theta + \omega_A \omega_B \cos \theta) + I \omega_B \cos \theta \cdot \omega_A = [M_{B,2}]_{\mathcal{M}} \\ I \cdot (\dot{\omega}_A \cos \theta - \omega_A \omega_B \sin \theta) - I \omega_B \sin \theta \cdot \omega_A = [M_{B,3}]_{\mathcal{M}} \end{cases}$$